



Tennessee Valley Authority, Post Office Box 2000, Decatur, Alabama 35609-2000

November 9, 2017

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

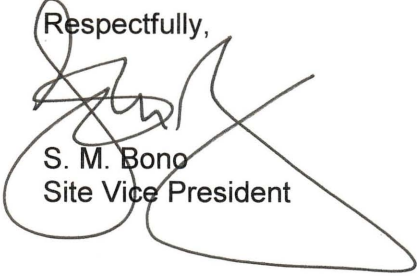
Browns Ferry Nuclear Plant, Units 1, 2, and 3
Renewed Facility Operating License Nos. DPR-33, DPR-52, and DPR-68
NRC Docket Nos. 50-259, 50-260, and 50-296

Subject: **Licensee Event Report 50-259/2017-004-00**

The enclosed Licensee Event Report provides details of the concurrent inoperability of both trains of the Unit 1 and 2 Control Bay Chillers, and the safety systems which they supported. The Tennessee Valley Authority is submitting this report in accordance with 10 CFR 50.73(a)(2)(v)(A)/(B)/(C)/(D), as any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to shut down the reactor and maintain it in a safe shutdown condition, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact J. L. Paul, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,



S. M. Bono
Site Vice President

Enclosure: Licensee Event Report 50-259/2017-004-00 - Safety System Functional Failures Due to the Inoperability of Both Unit 1 and 2 Control Bay Chillers

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant

ENCLOSURE

**Browns Ferry Nuclear Plant
Units 1, 2, and 3**

Licensee Event Report 50-259/2017-004-00

Safety System Functional Failures Due to the Inoperability of Both Unit 1 and 2 Control Bay Chillers

See Enclosed

NRC FORM 366 (04-2017)		U.S. NUCLEAR REGULATORY COMMISSION			APPROVED BY OMB: NO. 3150-0104			EXPIRES: 03/31/2020			
 LICENSEE EVENT REPORT (LER)					Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.						
1. FACILITY NAME Browns Ferry Nuclear Plant, Unit 1					2. DOCKET NUMBER 05000259			3. PAGE 1 OF 10			
4. TITLE Safety System Functional Failures Due to the Inoperability of Both Unit 1 and 2 Control Bay Chillers											
5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED		
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER	
09	10	2017	2017	- 004	- 00	11	09	2017	Browns Ferry Nuclear, Unit 2	05000260	
									FACILITY NAME	DOCKET NUMBER	
									Browns Ferry Nuclear, Unit 3	05000296	
9. OPERATING MODE		11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)									
1		☐ 20.2201(b)			☐ 20.2203(a)(3)(i)			☐ 50.73(a)(2)(ii)(A)		☐ 50.73(a)(2)(viii)(A)	
		☐ 20.2201(d)			☐ 20.2203(a)(3)(ii)			☐ 50.73(a)(2)(ii)(B)		☐ 50.73(a)(2)(viii)(B)	
		☐ 20.2203(a)(1)			☐ 20.2203(a)(4)			☐ 50.73(a)(2)(iii)		☐ 50.73(a)(2)(ix)(A)	
		☐ 20.2203(a)(2)(i)			☐ 50.36(c)(1)(i)(A)			☐ 50.73(a)(2)(iv)(A)		☐ 50.73(a)(2)(x)	
100		☐ 20.2203(a)(2)(ii)			☐ 50.36(c)(1)(ii)(A)			☒ 50.73(a)(2)(v)(A)		☐ 73.71(a)(4)	
		☐ 20.2203(a)(2)(iii)			☐ 50.36(c)(2)			☒ 50.73(a)(2)(v)(B)		☐ 73.71(a)(5)	
		☐ 20.2203(a)(2)(iv)			☐ 50.46(a)(3)(ii)			☒ 50.73(a)(2)(v)(C)		☐ 73.77(a)(1)	
		☐ 20.2203(a)(2)(v)			☐ 50.73(a)(2)(i)(A)			☒ 50.73(a)(2)(v)(D)		☐ 73.77(a)(2)(i)	
		☐ 20.2203(a)(2)(vi)			☐ 50.73(a)(2)(i)(B)			☐ 50.73(a)(2)(vii)		☐ 73.77(a)(2)(ii)	
		☐ 50.73(a)(2)(i)(C)						☐ OTHER		Specify in Abstract below or in NRC Form 366A	
12. LICENSEE CONTACT FOR THIS LER											
LICENSEE CONTACT Ryan Coons, Licensing Engineer								TELEPHONE NUMBER (Include Area Code) 256-729-2070			
13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT											
CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX		
B	KM	TE	TR04	Y	B	KM	CAP	N990	Y		
14. SUPPLEMENTAL REPORT EXPECTED								15. EXPECTED SUBMISSION DATE			
☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO								MONTH	DAY	YEAR	
								N/A	N/A	N/A	
ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines) On September 9, 2017, at 1520 Central Daylight Time (CDT), Operations personnel declared Unit 1/2 Control Bay (CB) Chiller (CHLR) A inoperable after observing that its outlet water temperature exceeded its specified limits. On September 10, 2017, at 1151 CDT, the Unit 1/2 CB CHLR B was declared inoperable after failing to start and providing a flashing error code to an operator in the field. The concurrent inoperability of both Unit 1/2 CB CHLRs required Browns Ferry Nuclear Plant (BFN) to declare the immediate inoperability of the electrical equipment in the affected electric board rooms, including all four Unit 1/2 4160 V Shutdown Boards, resulting in a Safety System Functional Failure, and entry into Technical Specifications (TS) Limiting Conditions for Operation (LCO) 3.0.3 for BFN, Units 1, 2, and 3. On September 10, 2017, at 2110 CDT, the Unit 1/2 CB CHLR B was restored to operable status following a capacitor replacement, and LCO 3.0.3 was exited. The Unit 1/2 CB CHLR A failure was caused by a thermistor failure, likely due to a break in the cable system. The Unit 1/2 CB CHLR B failure was due to a capacitor failing to hold its charge. Corrective actions include changing the preventive maintenance frequency interval for CB CHLR capacitor replacement from 10 years to 4 years. In addition, the concurrent Unit 1/2 CB CHLR failures were caused by a design which lacked defense-in-depth, where removing one Unit 1/2 CB CHLR from service for unplanned corrective maintenance created a single-point failure risk. A design change has been proposed to install a cross-tie between the Unit 3 Chillers and the Unit 1/2 Chillers to provide cooling in the event that Unit 1/2 CB CHLR A and B were to both become inoperable.											

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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		YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 1	05000-259	2017	- 004	- 00

NARRATIVE**I. Plant Operating Conditions Before the Event**

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Units 1, 2, and 3, were in Mode 1 at 100 percent power.

II. Description of Event**A. Event Summary**

On September 9, 2017, at 1520 Central Daylight Time (CDT), Operations personnel observed that the outlet water temperature on Unit 1/2 Control Bay (CB) Chiller (CHLR) [CHU] A exceeded its specified limits. Unit 1/2 CB CHLR A was then declared inoperable for troubleshooting on September 9, 2017. The inoperability of Unit 1/2 CB CHLR A required BFN, Units 1 and 2, to enter Technical Specification (TS) Limiting Condition for Operation (LCO) 3.7.4, Control Room Air Conditioning (AC) System, Condition A.

On September 10, 2017, at 1151 CDT, Unit 1/2 CB CHLR B failed to start, and an operator in the field noted a flashing error code. Operations personnel noted that the A Compressor on Unit 1/2 CB CHLR B was not running, and was not warm to the touch. Operations personnel declared Unit 1/2 CB CHLR B inoperable at this time.

The concurrent inoperability of both Unit 1/2 CB CHLRs required Browns Ferry Nuclear Plant (BFN) to declare the immediate inoperability of the electrical equipment in the affected electric board rooms, including all four Unit 1/2 4160 V (4kV) Shutdown Boards (SD BDs) [ECBD]. This resulted in multiple TS LCO and Technical Requirements Manual (TRM) LCO entries and a Safety System Functional Failure for safety systems on all three Units.

The concurrent inoperability of Unit 1/2 CB CHLRs A and B also required BFN, Units 1 and 2, to enter TRM LCO 3.7.6, Electric Board Room Air Conditioning (AC) System, Condition B. This LCO requires the unit, with two Unit 1/2 electric board room AC subsystems inoperable, to declare the electrical equipment in the electric board rooms inoperable immediately.

In particular, this event resulted in the loss of two trains of Standby Gas Treatment (SGT) [BH], which required entry into TS LCO 3.0.3 for BFN, Units 1, 2, and 3. As a result of the unplanned entry into TS LCO 3.0.3, all three BFN Units were required to be in Mode 2 by 2151 CDT on September 10, 2017. The NRC Headquarters Office granted BFN an Enforcement Discretion from complying with TS LCO 3.0.3, Condition A, which extended the time to shutdown by 12 hours and 9 minutes, such that all three Units were now required to either exit TS LCO 3.0.3 or be in Mode 2 by 1000 CDT on September 11, 2017.

On September 10, 2017 at 1945 CDT, the Tennessee Valley Authority (TVA) submitted an 8-hour event notification (EN) to the NRC in accordance with 10 CFR 50.72(b)(3)(v)(A)/(B)/(C)/(D), as any event or condition that could have prevented the fulfillment of the safety function of structures or



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systems that are needed to shut down the reactor and maintain it in a safe shutdown condition, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident.

On September 10, 2017, at 2110 CDT, Unit 1/2 CB CHLR B was declared operable, and BFN, Units 1, 2, and 3 exited all applicable TS LCOs and the Notice of Enforcement Discretion (NOED). Unit 1/2 CB CHLR A was declared operable on September 13, 2017 at 1445 CDT following the completion of repairs and post-maintenance testing.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

At the start of this event, Unit 1/2 CB CHLR A was inoperable for unplanned corrective maintenance due to a failed thermistor. This resulted in the concurrent inoperability of Unit 1/2 CB CHLR A and B.

C. Dates and approximate times of occurrences

Dates and Times

Occurrence

September 9, 2017
at 1520 CDT

The Control Bay Auxiliary Unit Operator (AUO) noted Unit 1/2 CB CHLR A alarms for "Chiller A Trouble" and "Chilled Water High Temp". An additional AUO also reported a flashing error code. Operations personnel declared CB CHLR A inoperable, and Units 1 and 2 entered TS LCO 3.7.4, Condition A. The chiller remained in-service pending troubleshooting.

September 9, 2017
at 1715 CDT

The CB CHLR system engineer notified Operations personnel that the B circuit for the Unit 1/2 CB CHLR A failed, potentially due to a failed thermistor [TE]. The Unit 1/2CB CHLR A continued running at half load.

September 10, 2017
at 1130 CDT

Operations personnel alternated the Unit 1/2 CB CHLRs from A to B, in accordance with the applicable operating instruction.

September 10, 2017
at 1151 CDT

Unit 1/2 CB CHLR B was declared inoperable after an operator noticed it gave a flashing error code on its A Compressor, which did not appear to be running and was not warm to the touch. Since Unit 1/2 CB CHLR A was also inoperable, and TS LCO 3.7.4 Condition A was previously entered on Unit 1/2 for this condition, Operations personnel entered all applicable TRM LCO and TS LCOs at this time

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Dates and Times

September 10, 2017
at 1746 CDT

Occurrence

NRC Headquarters Office granted BFN Enforcement Discretion from complying with the TS LCO 3.8.7 required action to enter TS LCO 3.0.3 Condition A, which would have required all Units to be in Mode 2 by 2151 CDT on September 10, 2017. The Enforcement Discretion extended this time by 12 hours and 9 minutes, such that all three Units were now required to either exit TS 3.0.3 or be in Mode 2 by 1000 CDT on September 11, 2017.

September 10, 2017
at 1945 CDT

Operations personnel issued an 8-hour EN 52959 to the NRC.

September 10, 2017
at 2110 CDT

Unit 1/2 CB CHLR B was declared operable after being placed back in service with one new capacitor [CAP] and one degraded capacitor. Unit 1/2 CB CHLR B started and all parameters were monitored and verified to be correct.

All applicable Units 1, 2, and 3 TS and TRM LCOs were exited, including TS LCO 3.0.3, therefore, exiting the NOED.

September 13, 2017
at 1445 CDT

Unit 1/2 CB CHLR A was declared operable after the completion of repairs and post-maintenance testing.

D. Manufacturer and model number of each component that failed during the event

The thermistor temperature element was manufactured by Trane Air Conditioning, with vendor part number SEN00307.

The failed capacitor was a metalized polypropylene quick-connect 15 microFarad capacitor, manufactured by Ronken Industries, Inc. and qualified by Nuclear Logistics, Inc. with manufacturer part number P91C23156H05.

E. Other systems or secondary functions affected

Loss of the Unit 1/2 CB CHLRs resulted in a loss of the Unit 1 and 2 Control Room AC System. Additionally, the inoperability of all BFN, Unit 1 and 2, 4kV SD BDs rendered the following systems inoperable:



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4kV SD BD A	4kV SD BD B	4kV SD BD C	4kV SD BD D
Emergency Diesel Generator (EDG) A [DG]	EDG B	EDG C	EDG D
Residual Heat Removal (RHR)[BO] Pump [P] 1A RHR Pump 2A	RHR Pump 1C RHR Pump 2C	RHR Pump 1B RHR Pump 2B	RHR Pump 1D RHR Pump 2D
Core Spray (CS) [BM] Pump 1A CS Pump 2A	CS Pump 1C CS Pump 2C	CS Pump 1B CS Pump 2B	CS Pump 1D CS Pump 2D
RHR Service Water (RHRSW) [CC] Pump A1 RHRSW Pump A2	RHRSW Pump C1 RHRSW Pump C2	RHRSW Pump B2 RHRSW Pump B3 (EECW) [BI]	RHRSW Pump D2 RHRSW Pump D3 (EECW)
Standby Liquid Control (SLC) [BR] Pump 1A	SLC Pump 2A	SLC Pump 1B	SLC Pump 2B
SGT A	CB CHLR 3A		SGT B
Control Room Emergency Ventilation System (CREVS) A [UF]			

F. Method of discovery of each component or system failure or procedural error

An AUO noted Unit 1/2 CB CHLR A alarms for "Chiller A Trouble" and "Chilled Water High Temp", and an additional AUO also reported a flashing error code.

Unit 1/2 CB CHLR B was declared inoperable after an operator noticed it gave a flashing error code on it's A Compressor, which did not appear to be running and was not warm to the touch.

G. The failure mode, mechanism, and effect of each failed component

For Unit 1/2 CB CHLR B, the capacitor failed to hold its charge.

For Unit 1/2 CB CHLR A, post-event testing indicated no failure mode for the thermistor, which continued to operate as-designed with no deviations from the manufacturer's tolerance specifications. As such, this failure is attributed to a loose or open cable connection between the system and the thermistor.

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H. Operator actions

In response to identified error codes on Unit 1/2 CB CHLR A and B, Operations personnel declared Unit 1/2 CB CHLR A and B inoperable. Additionally, Operations personnel declared CREVS A, SGT A and B, and EDG A, B, C, and D inoperable and entered TS LCO 3.0.3.

I. Automatically and manually initiated safety system responses

There were no automatic or manual safety system responses associated with this condition.

III. Cause of the event**A. Cause of each component or system failure or personnel error**

The cause of the concurrent Unit 1/2 CB CHLR failures was a design which lacked defense-in-depth. Specifically, removing one Unit 1/2 CB CHLR from service for unplanned corrective maintenance creates a single-point failure risk in the event that the second Unit 1/2 CB CHLR becomes inoperable.

The most likely cause of the Unit 1/2 CB CHLR A thermistor failure was a loose or open connection in the cable system on the system side of the thermistor. This cable was cut, removed, and replaced during maintenance as part of the corrective maintenance to replace the failed thermistor for the current event. The open connection was likely due to a broken cable, but since the cable was not preserved for testing, this cannot be confirmed or denied. Given the age of this cable, the likely failure mode was infant mortality.

The cause of the Unit 1/2 CB CHLR B failure was that its capacitors in line with the stepdown transformer had lost their ability to hold a high enough voltage during chiller startup to maintain power to its controller. This was first evidenced by improper controller operation and the multiple error codes observed by maintenance personnel during troubleshooting and restart attempts. This was verified by measuring stepped down line voltage, which was found to be below specifications, indicating either a capacitor issue or a transformer problem.

B. Cause(s) and circumstances for each human performance related root cause

There were no human performance related root causes associated with this event.

IV. Analysis of the event

This event resulted from a capacitor failure on the Unit 1/2 CB CHLR B train when the redundant Unit 1/2 CB CHLR A train was already inoperable for maintenance. Equipment in electric board rooms rely on chilled water from this system for proper functioning in all conditions, including accident conditions. Therefore, all Unit 1 and 2 4kV SD BDs and associated EDGs were declared inoperable along with the chiller.



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The safety function of the 4kV SD BDs, as part of the Standby AC Power System, is to provide a self-contained, highly reliable source of power, as required for the Engineered Safeguards System, so that no single credible event can disable the core standby cooling functions or their supporting auxiliaries. Inoperability of the Unit 1 and 2 4kV SD BDs and EDGs caused the loss of several safety systems and the unplanned entry into a shutdown LCO for BFN Units 1 and 2. The loss of two trains of SGT also caused the unplanned entry into a Unit 3 shutdown.

The inoperability of Unit 1/2 CB CHLR A required BFN, Units 1 and 2, to enter TS LCO 3.7.4, Control Room Air Conditioning (AC) System, Condition A, which, when the Units are in Modes 1, 2, or 3, requires the affected Units 1 and 2 control room AC subsystem to be restored to operable status within 30 days.

On September 10, 2017, at 1151 CDT, Unit 1/2 CB CHLR B failed to start, and an operator in the field noted a flashing error code. Operations personnel noted that the A Compressor on Unit 1/2 CB CHLR B was not running, and was not warm to the touch. Operations personnel declared Unit 1/2 CB CHLR B inoperable at this time.

The concurrent inoperability of Unit 1/2 CB CHLRs A and B required BFN, Units 1 and 2, to enter TS LCO 3.7.4, Control Room Air Conditioning (AC) System, Condition B, which requires an immediate action to restore one Unit 1 and 2 control room AC subsystem to operable status; placing an alternate cooling method in operation within 24 hours; and restoring operability to one control room AC subsystem within 7 days.

The concurrent inoperability of Unit 1/2 CB CHLRs A and B also required BFN, Units 1 and 2, to enter Technical Requirements Manual (TRM) LCO 3.7.6, Electric Board Room Air Conditioning (AC) System, Condition B. This LCO requires the unit, with two Unit 1/2 electric board room AC subsystems inoperable, to declare the electrical equipment in the electric board rooms inoperable immediately.

For BFN, Units 1 and 2, all four (4kV) SD BDs were declared inoperable.

With 4kV SD BDs A, B, C, and D inoperable, BFN, Units 1 and 2, entered TS LCO 3.8.7, Distribution Systems – Operating, Condition A, which requires that, with one Unit 1 and 2 4kV SD BD inoperable, the Unit 1 and 2 4kV SD BD be restored to operable status within five days, and that the associated EDG be declared inoperable immediately. BFN, Units 1 and 2, declared EDG A, EDG B, EDG C, and EDG D inoperable.

With 4kV SD BDs A, B, C, and D inoperable, BFN, Units 1, 2, and 3 entered TS LCO 3.8.7, Conditions F (Unit 1) and G (Units 2 and 3). These LCOs require, with one or more required other units AC boards inoperable, that the affected SGT or CREVS subsystems be declared inoperable immediately. BFN, Units 1 and 2, declared SGT A, SGT B, and CREVS A inoperable and entered TS LCO 3.8.7, Conditions H (Unit 1) and I (Unit 2). These LCOs require, with two or more electrical power distribution subsystems inoperable that result in a loss of function, that the unit enter TS LCO 3.0.3 immediately.



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With SGT A and B inoperable, BFN, Units 1, 2, and 3, entered TS LCO 3.6.4.3, SGT System, Condition D. This LCO requires immediate TS LCO 3.0.3 entry when two or three SGT subsystems are inoperable in Mode 1, 2, or 3.

With EDG A, EDG B, EDG C, and EDG D inoperable, BFN, Units 1 and 2, entered TS LCO 3.8.1, AC Sources - Operating, Condition B, for each inoperable EDG. Failure to meet any of the TS LCO 3.8.1, Condition B completion times requires entry into TS LCO 3.8.1, Condition I, which requires Units 1 and 2 to be in Mode 3 within 12 hours and Mode 4 within 36 hours.

Additionally, with EDG A, EDG B, EDG C, and EDG D inoperable, BFN, Unit 3 entered TS LCO 3.8.1, Condition K, for each inoperable EDG. This LCO requires declaring the required feature(s) supported by the inoperable Unit 1 and 2 EDG inoperable when the redundant required feature(s) are inoperable, within 4 hours from the discovery of Condition K concurrent with inoperability of redundant required feature(s); and declaring the affected SGT and CREVs subsystem(s) inoperable within 30 days.

With CREVS A inoperable, BFN, Units 1, 2, and 3, entered TS LCO 3.7.3, Control Room Emergency Ventilation (CREV) System, Condition A. This LCO requires an inoperable CREV subsystem to be restored to operable status within 7 days, when the unit is in Modes 1, 2, or 3, and during operations with a potential for draining the reactor vessel (OPDRVs).

V. Assessment of Safety Consequences

This event resulted in inoperability of all four Unit 1 and 2 4kV SD BDs and their associated EDGs; however, the equipment remained available during the event. This equipment was declared inoperable because it was unable to perform its safety function under all design basis conditions for supplying reliable power to safety-related equipment in order to shut down the reactor and maintain it in a safe shutdown condition, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident.

The 4kV SD BDs are located in the Unit 1 and 2 Electric Board Rooms (EBR). System failure could result if the equipment operating temperature in the EBRs exceeded the 104 degrees Fahrenheit (F) operability limit. During this event, non-qualified temporary cooling was available to ensure the temperature in the EBR would not exceed the 104 degrees F operability limit of the associated safety-related components. Therefore, there was no significant risk to the health and safety of the public or plant personnel for this event.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

Non-qualified temporary cooling was available to ensure that the EBR room temperatures would not exceed the 104 degrees F operability limit of the associated safety-related components.



LICENSEE EVENT REPORT (LER) CONTINUATION SHEET

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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B. For events that occurred when the reactor was shut down, availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

This condition did not occur during a shutdown.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service

This event resulted in inoperability of multiple plant safety systems, from September 10, 2017 at 1151 CDT, when the Unit 1/2 CB CHLR B was declared inoperable after a flashing error code was discovered on the A Compressor, until September 10, 2017 at 2110 CDT when the Unit 1/2 CB CHLR B was declared operable after verification that a capacitor replacement corrected the error.

VI. Corrective Actions

Corrective Actions are being managed by TVA's corrective action program under Condition Reports (CRs) 1336821 and 1340251.

A. Immediate Corrective Actions

Maintenance crews worked in parallel to restore both Unit 1/2 CB CHLR trains to operable status. Unit 1/2 CB CHLR B was returned to service first, after replacing a 15 microFarad capacitor and reinstalling a degraded 20 microFarad capacitor that tested at 14 microFarads. The chiller was started and all parameters were monitored and verified as satisfactory. Unit 1/2 CB CHLR A was later declared operable after replacement of the thermistor and completion of its associated post-maintenance testing.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future

On September 13, 2017 the CB CHLR A capacitors were replaced as a precautionary measure. The degraded CB CHLR B capacitor was replaced with a new capacitor on September 14, 2017.

The following corrective actions to prevent recurrence are currently are in progress:

1. An action was initiated to ensure that outstanding CB CHRL work orders are scheduled. Another action is in place to ensure the completion of these work orders.
2. The preventive maintenance frequency interval for CB CHLR capacitor replacement will be changed from 10 years to 4 years.
3. A design change has been planned to prevent future recurrence by installing a cross-tie between the Unit 3 Chillers and the Unit 1/2 Chillers. This cross-tie of chilled water between Unit 3 and Units 1/2 would provide cooling in the event that both Unit 1/2 CB CHLRs A and B were to become inoperable.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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VII. Previous Similar Events at the Same Site

A similar event occurred at BFN on July 14, 2015, when the Unit 1 and 2 CB CHLR A was rendered inoperable when its condenser coils became fouled with dirt and insects while Unit 1/2 CB CHLR B was rendered inoperable for scheduled maintenance. This event was recorded in Licensee Event Report 05000259/2015-003-00.

The corrective actions to prevent the recurrence of the July 2015 event were focused on inspecting the condition of equipment for the effects of dirt, insects, and other natural phenomena, and would not have prevented this event.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.